

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 23

Hypertension and Heart Failure

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Recommended Reading

- Rhoades, R. & Bell, D. (2012); *Medical Physiology: Principles for Clinical Medicine*, 5th Edition; Wolters Kluwer. Chapter 11

Clinical cases:

Adapted from Cases and Problems in Physiology

Practice Questions: BRS, Guyton, Pretest

*DLA on control of blood pressure.

Lecture Objectives

Lecture Objective	
SOM.MK.II.BPM1.3.CPR.2. PHYS.0179	Define and classify hypertension <i>What is hypertension? Definition, Classification, Risk factors, Features</i>
SOM.MK.II.BPM1.3.CPR.2. PHYS.0180	Define chronic congestive cardiac failure <i>What are long term consequences of Hypertension?</i>
SOM.MK.II.BPM1.3.CPR.2. PHYS.0181	Describe the compensatory responses to mild cardiac failure and explain the outcomes of moderate to severe cardiac failure. <i>What are the compensatory mechanisms to these long term changes?</i>
SOM.MK.II.BPM1.3.CPR.2. PHYS.0182	Describe the major cardiovascular changes that occur with ageing (i.e. compliance changes) <i>Is hypertension a sequelae of old age?</i>

Definition of Hypertension – JNC 8

Hypertension is “that level of BP above which investigation and treatment do more good than harm”

	Normal	Elevated	Hypertension Stage 1	Hypertension Stage 2	Hypertensive Crisis
 Systolic blood pressure (mm Hg)	< 120	120-129	130-139	≥ 140	> 180
	and	and	or	or	and/or
 Diastolic blood pressure (mm Hg)	< 80	< 80	80-89	≥ 90	> 120

90% of cases – cause unknown, slow progression, asymptomatic for years,

Small % *hypertensive urgency* or *hypertensive emergency*, rapidly progressing, Medical emergency, may or may not have severe organ damage risk (e.g., kidneys, brain)

Risk factors: age, obesity, diabetes, physical inactivity, excess salt intake, excess alcohol intake, family history; African American

Causes of secondary hypertension

Only 10% of cases – specific disease or abnormality - Secondary

Condition associated with Secondary Hypertension

Mechanism

Renal Artery Stenosis

Decreased blood flow triggers RAS activation → Increased Angiotensin II → Vasoconstriction and higher BP

Renal Disease

Reduced sodium and water excretion → Increased blood volume → Elevated BP

Conn's Syndrome (Hyperaldosteronism)

Excess aldosterone → Salt and water retention → Increased BP

Pheochromocytoma

Excess epinephrine secretion → Increased heart rate and vasoconstriction → Higher BP

Pre-eclampsia Toxemia

Pregnancy-related condition → Increased blood pressure

Hyperthyroidism

Increased heart rate, systemic vascular resistance, and cardiac output → Higher systolic BP

Cushing's Disease/Syndrome

Overactivation of renin-angiotensin system → Increased BP

Pathophysiology of Hypertension

No unifying hypothesis, Multiple defects of blood pressure regulation interacting with environmental stressors (multifactorial)

$$\text{MAP} = \text{CO} \times \text{TPR}$$

1° Pathology: $\uparrow$ TPR or $\uparrow$ CO And/or $\uparrow$ blood volume

Pump-based Hypertension: $\uparrow$ CO

*Occurs in younger patients, amenable to beta blockers**
(overdeveloped alerting response; excessive Sympathetic effects on heart)

Vascular Resistance-based Hypertension: $\uparrow$ TPR

Occurs more in older patients. (Smooth muscle abnormally sensitive to vasoconstrictors, endothelial cell dysfunction – abnormal regulation of vascular tone by local factors e.g. NO)

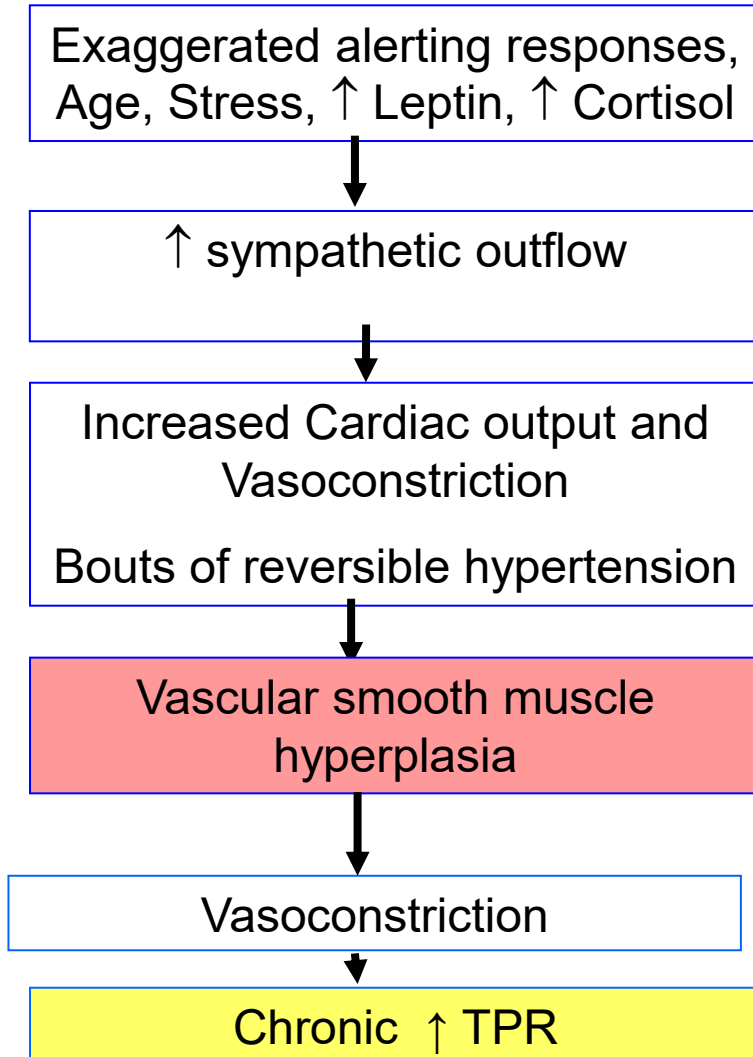
Volume-based Hypertension: $\uparrow$ retention of Na and H₂O

Renal parenchymal disease, renovascular disorders.
(Failure of the Renin Angiotensin System to regulate BP)

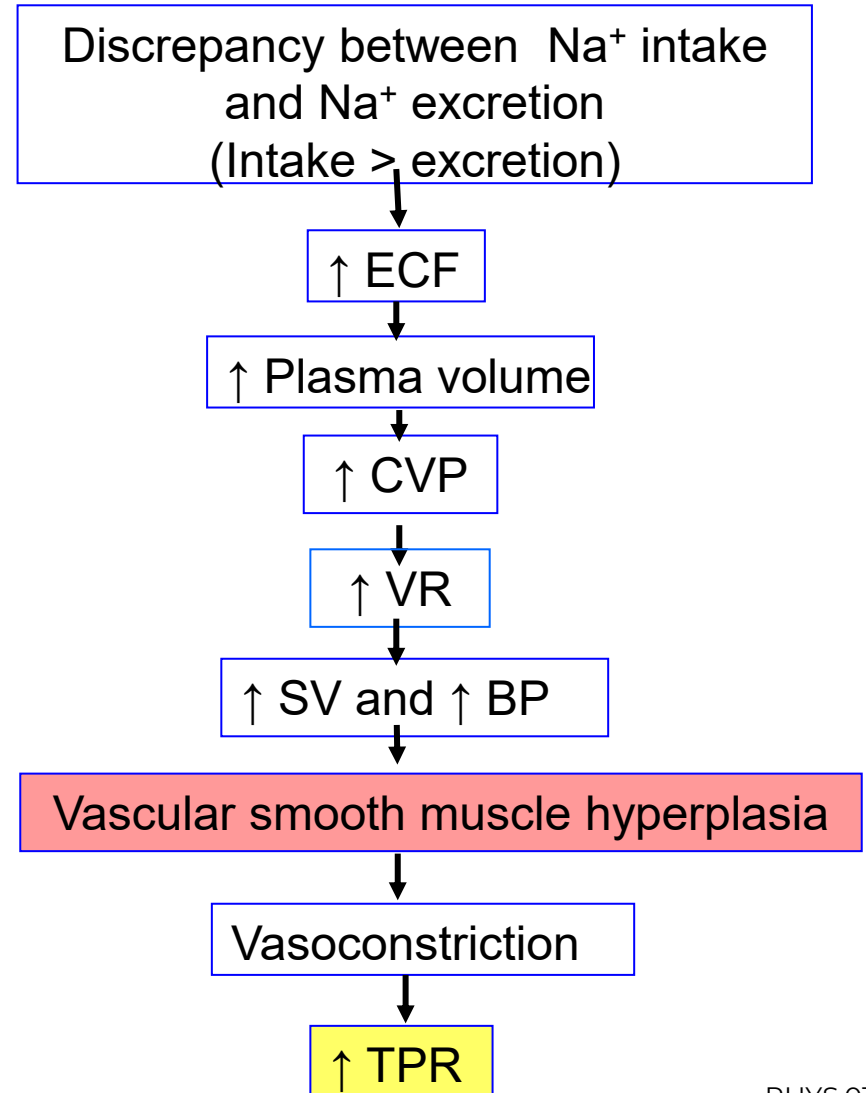
Note: normally, no matter how high the CO or TPR, renal excretion has the capacity to return BP to normal by $\downarrow$ blood volume. Maintenance of chronic hypertension involves kidneys.

Mechanism causing Hypertension

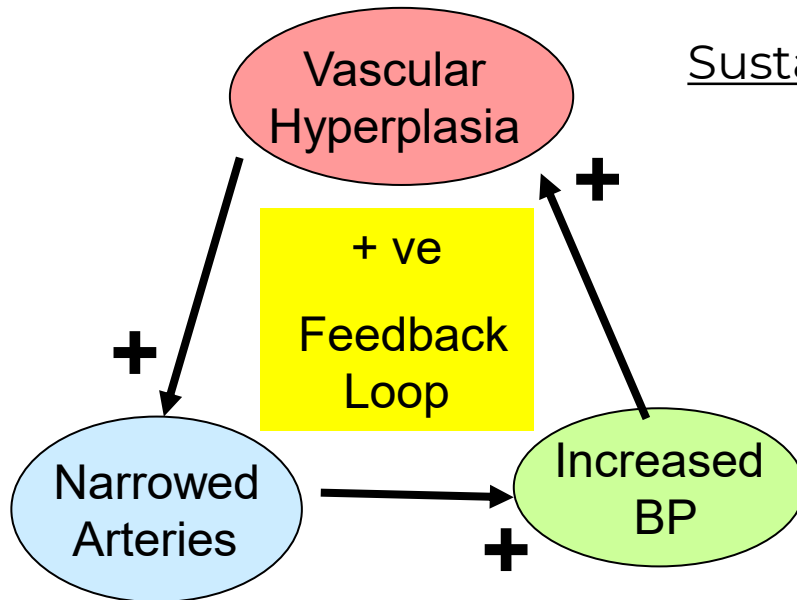
Neurogenic or Stress



Salt imbalance or Renal Imbalance(Volume based)



Sustained blood pressure creates a positive-feedback loop.

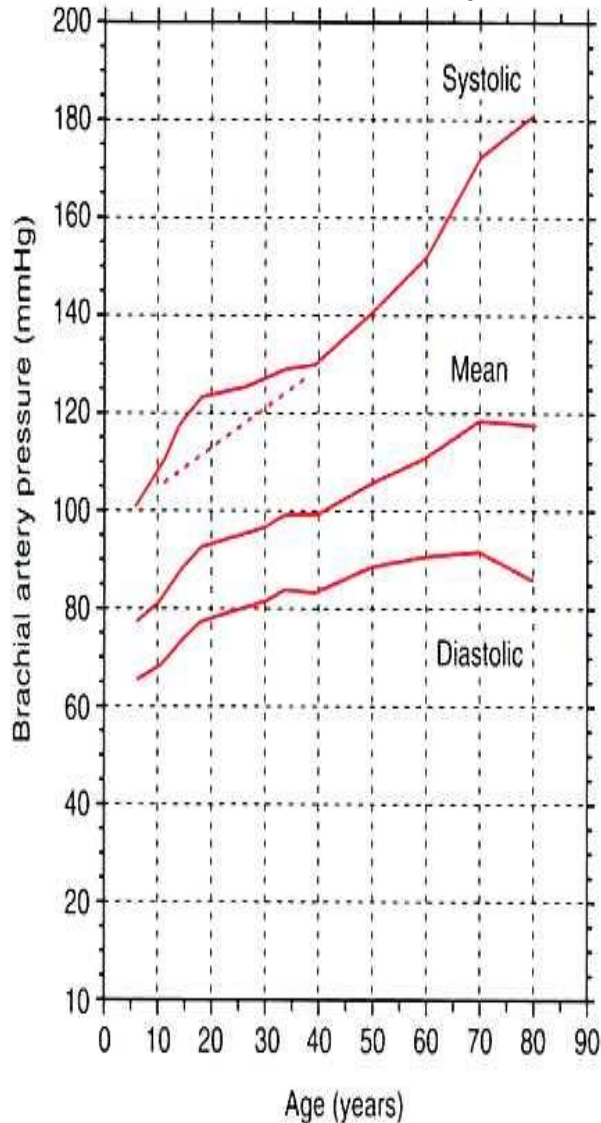


Sustained pump activity + sustained vascular resistance
= Sustained Blood Pressure

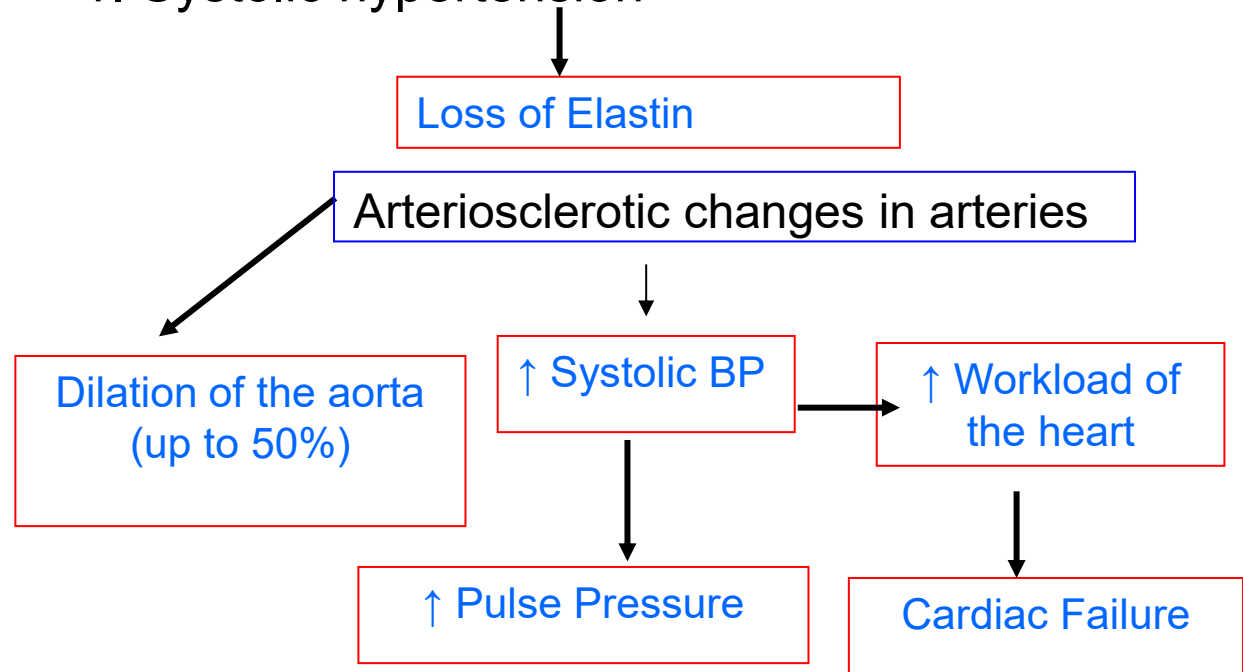
- ↑ bouts of high blood pressure
- Leads to vascular hypertrophy
- Leads to vasoconstriction
- Further narrows arteries

Mechanism of age-related hypertension

Greater increase in Systolic BP



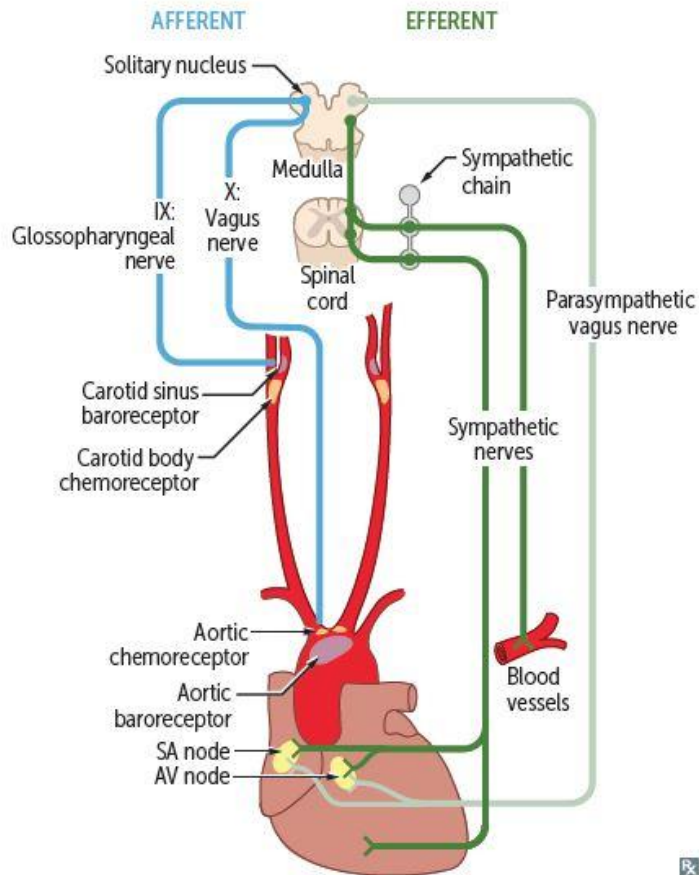
1. Systolic hypertension



2. ↓ Baroreceptor Sensitivity

3. ↓ Cardiac Performance in Exercise

Baroreceptors and chemoreceptors



Receptors:

- Aortic arch transmits via vagus nerve to solitary nucleus of medulla (responds to changes in BP).
- Carotid sinus (dilated region at carotid bifurcation) transmits via glossopharyngeal nerve to solitary nucleus of medulla (responds to changes in BP).

Baroreceptors:

- Hypotension—↓ arterial pressure → ↓ stretch → ↓ afferent baroreceptor firing → ↑ efferent sympathetic firing and ↓ efferent parasympathetic stimulation → vasoconstriction, ↑ HR, ↑ contractility, ↑ BP. Important in the response to severe hemorrhage.
- Carotid massage—↑ pressure on carotid sinus → ↑ stretch → ↑ afferent baroreceptor firing → ↑ AV node refractory period → ↓ HR.
- Component of Cushing reflex (triad of hypertension, bradycardia, and respiratory depression)—↑ intracranial pressure constricts arterioles → cerebral ischemia → ↑ $p\text{CO}_2$ and ↓ pH → central reflex sympathetic ↑ in perfusion pressure (hypertension) → ↑ stretch → peripheral reflex baroreceptor-induced bradycardia.

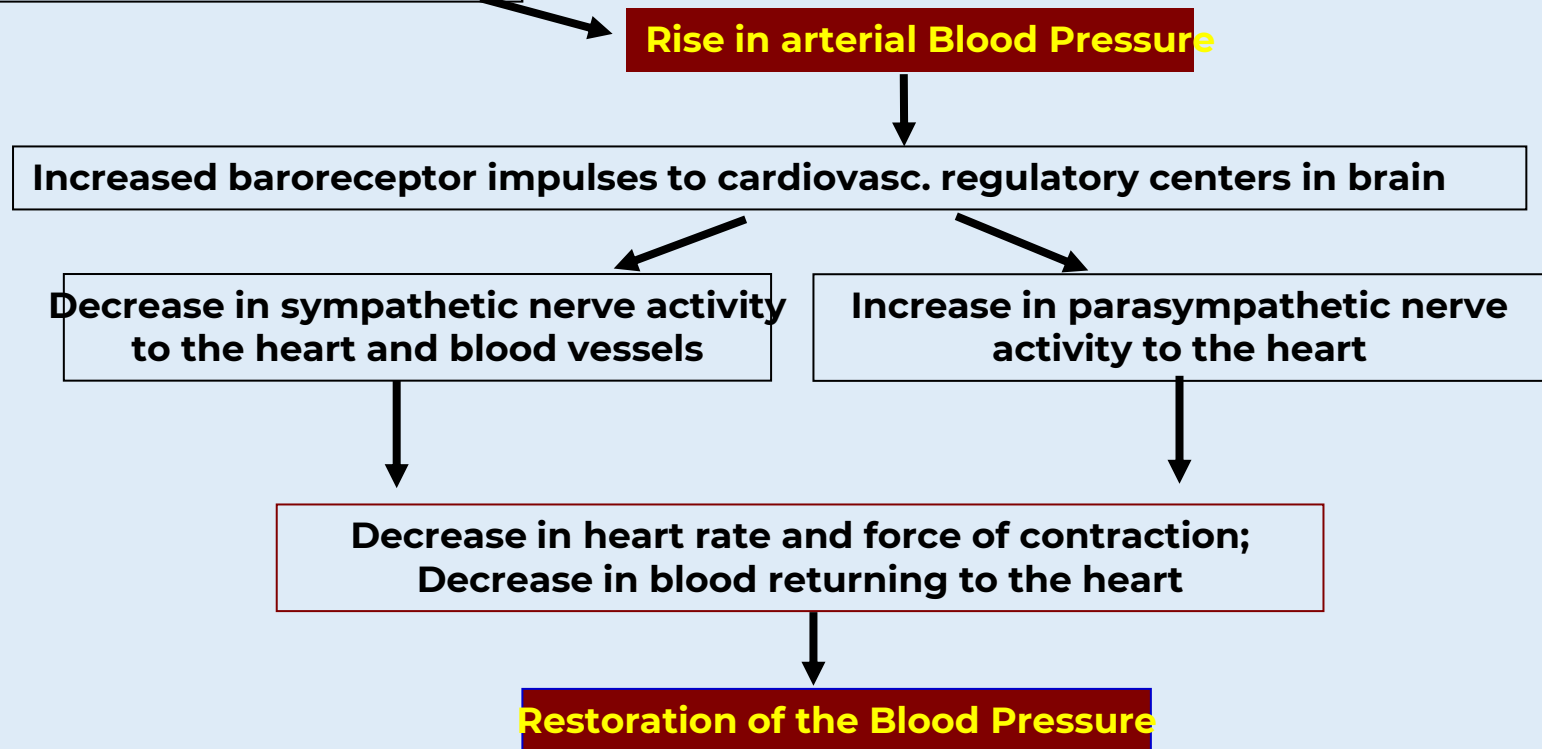
Chemoreceptors:



- Peripheral—carotid and aortic bodies are stimulated by ↑ Pco_2 , ↓ pH of blood, and ↓ Po_2 (< 60 mm Hg).
- Central—are stimulated by changes in pH and Pco_2 of brain interstitial fluid, which in turn are influenced by arterial CO_2 as H^+ cannot cross the blood-brain barrier. Do not directly respond to Po_2 . Central chemoreceptors become less responsive with chronically ↑ Pco_2 (eg, COPD) → ↑ dependence on peripheral chemoreceptors to detect ↓ O_2 to drive respiration.

Baroreflex response to a sudden rise in BP

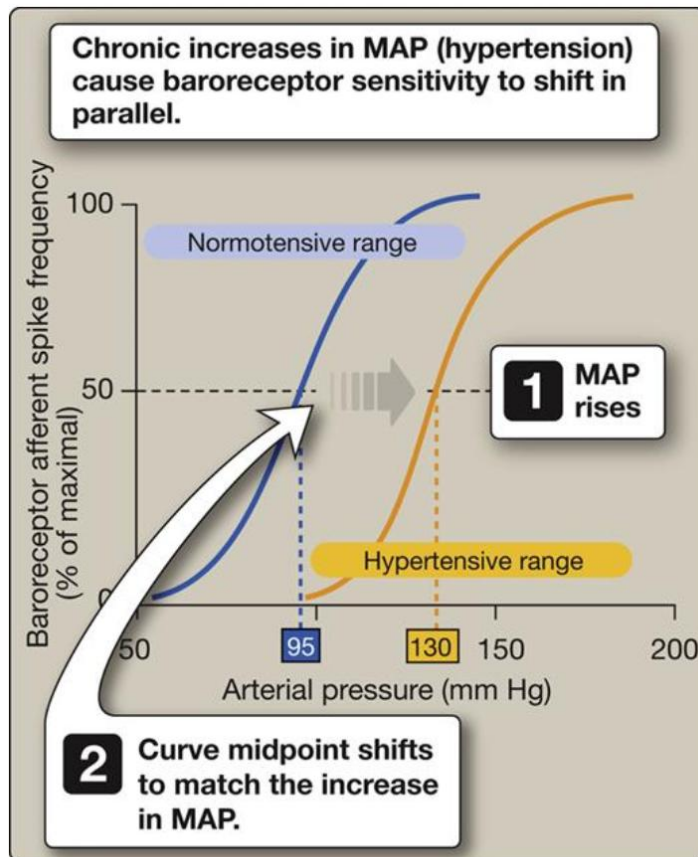
Example: Anxiety/Panic attack



Task: Draw a similar flow chart to show baroreflex response to a **sudden decrease** in the Blood Pressure

Role of Baroreceptors in sustained blood pressure

Baroreceptor response is immediate but short-lived



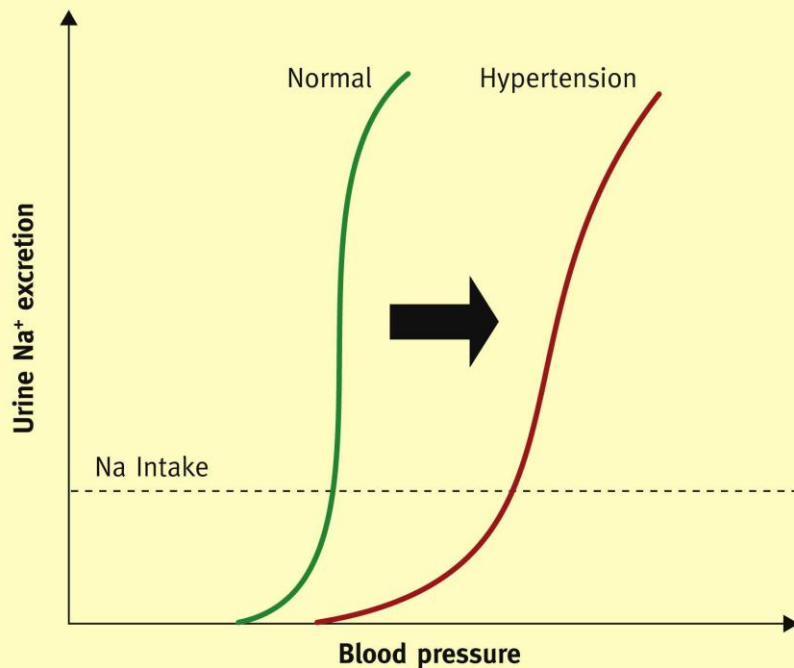
- **Baroreceptors modulate moment to moment changes in BP** – not involved in long term regulation
 - Baroreceptor Reflex does not prevent development of chronic HTN because baroreceptors constantly reset themselves.
 - Carotid sinus baroreceptor nerve firing rate and mean arterial pressure.
- ~ 1-2 days of high BP
- – baroreceptor firing (initially increased) goes back to normal (rapidly adapting receptors!)
 - Note: the baroreceptor reflex still operated in hypertension, but it operates around an elevated set point.

Role of kidney in sustained blood pressure

The Renin Angiotensin Aldosterone System response is delayed but lasts longer

- $\uparrow$ renal artery pressure leads to $\uparrow$ GFR and $\uparrow$ excretion of Na^+ and H_2O .

Pressure natriuresis and hypertension



From Smith T, Pinnock C, Lin T, Jones R. *Fundamentals of Anaesthesia*, 3rd Edn, 2009. With kind permission of Cambridge University Press.

- But in HTN, the pressure-natriuresis curve is shifted to right, shift of curve could be due to renal tissue damage or Renin Angiotensin System (RAS)
- In HTN, higher arterial pressure required to excrete Na^+ and H_2O ; retention resulting in $\uparrow$ blood volume
- However, sustained $\uparrow$ blood volume =
- $\uparrow$ sustained blood pressure

Summary... Hypertension

- Hypertension - persistent readings of $>140/90\text{mmHg}$
- Primary vs Secondary - causes
- $\text{MAP} = \text{CO} * \text{TPR}$
- Neurogenic vs renal or Stress vs salt imbalance $\rightarrow$ increasing the work of the heart, increasing the resistance or retaining blood volume
- Renal system is an important regulator
- Baroreceptor reflex helps but that cannot work for long...
- Treatment options based on pathogenesis theories

Why are we so worried about Hypertension?

Consequences of uncontrolled hypertension

↑ ^{ed} Work load of the LV	cardiac failure (congestive)
↑ ^{ed} risk of vascular disorders	cerebral hemorrhage aortic aneurysms
deposition of proteins → loss of nephrons	Renal damage and chronic renal failure
↑ ^{ed} risk of developing atheroma	coronary artery disease

(60% of CVAs and 50% of Ischemic Heart Disease
is due to inadequately treated hypertension).

Congestive Cardiac Failure

A pathophysiological state in which an abnormality of cardiac function is responsible for the failure of the heart to pump blood at a rate commensurate with the requirements of the metabolising tissue. *Braunwald 1980*

Clinical Features Do you remember the patient in the SIM Lab?

- Exercise Intolerance
- Breathlessness (Dyspnea), paroxysmal nocturnal dyspnea
- Fatigue
- Peripheral and pulmonary edema

Prevalence: 1% at age 50; 9% at age 80

400,000 new cases each year in US

Mortality: 50% die within 5 years of diagnosis (untreated)
Death is due to pump failure or arrhythmias

Most common diagnosis of hospitalized patients >65 years

Predisposing conditions: Chronic hypertension,
Coronary artery disease
Valvular heart disease

Classification of heart failure based on symptom severity and the amount of exertion needed to provoke symptoms. (Clinical)

_The New York Heart Association (NYHA) classification of heart failure is widely used in practice and in clinical studies. It follows:

Class I: No limitation of physical activity

Class II: Slight limitation of physical activity in which ordinary physical activity leads to fatigue, palpitation, dyspnea, or anginal pain; the person is comfortable at rest

Class III: Marked limitation of physical activity in which less-than-ordinary activity results in fatigue, palpitation, dyspnea, or anginal pain; the person is comfortable at rest

Class IV: Inability to carry on any physical activity without discomfort but also symptoms of heart failure or the anginal syndrome even at rest, with increased discomfort if any physical activity is undertaken

Heart failure may be the final and most severe manifestation of nearly every form of cardiac disease.

Heart failure is classified in several ways:

Anatomical - Left, right and biventricular

- Left – reduced left ventricular output, and or increased left atrial or pulmonary venous pressure – aortic stenosis leading to pulmonary congestion
- Right – reduced right ventricular output for any given right atrial pressure – lung disease – peripheral edema
- Biventricular – Failure of both ventricles either due to ischemia or progression of disease

**Functional - Diastolic and Systolic*

- Systolic - Impaired myocardial contraction
- Diastolic - Poor Ventricular filling or abnormal ventricular relaxation

Timeline - Acute and Chronic

- Acute – sudden e.g. MI
- Chronic – gradual impairment, progressive e.g. valvular heart disease

How do we know the heart is failing?

Cardiac Output = Stroke volume * Heart rate

1. **Stroke Volume** - influenced by:

- Contractility (decreased calcium uptake in the sarcoplasmic reticulum, low affinity of troponin for calcium, altered substrate metabolism from fatty acid to glucose oxidation, impaired energy production)
- Preload events (volume and pressure of blood in ventricles at **end diastole**)
- Afterload events (volume and pressure of blood in the ventricles **during systole**, resistance to blood leaving the heart)

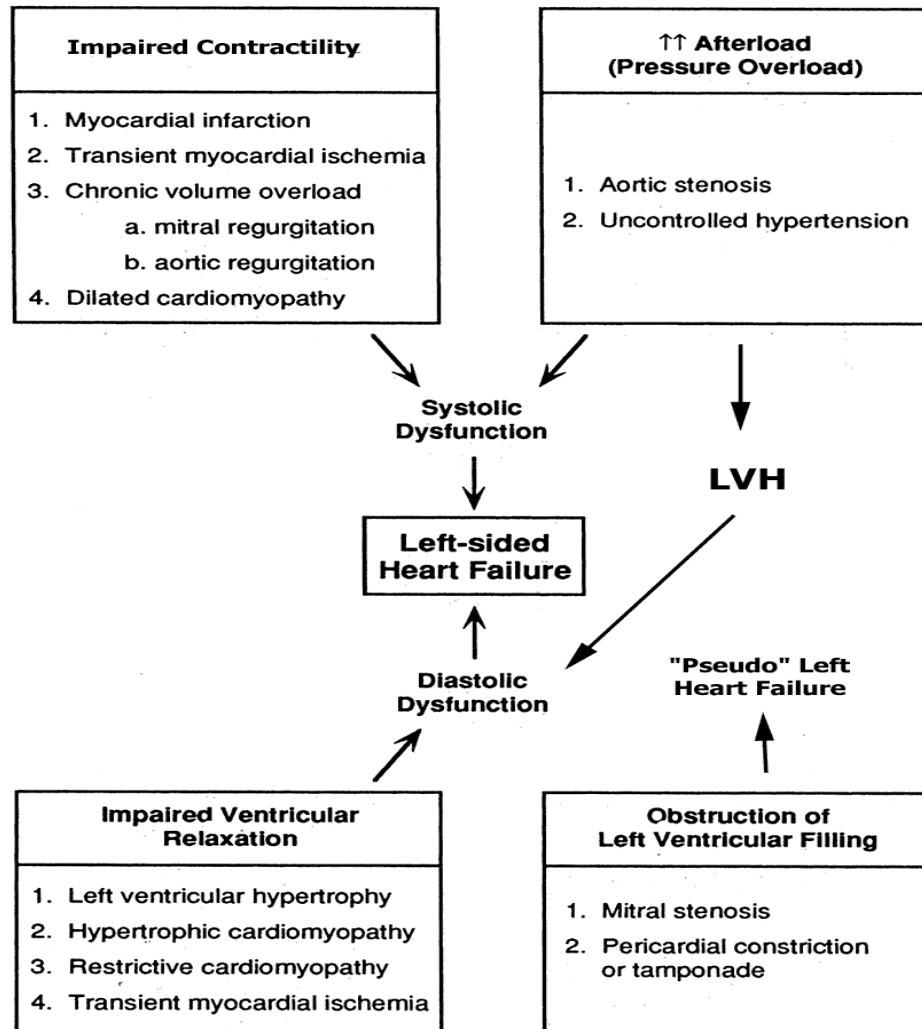
2. **Heart Rate**

A good indicator of contractility is the Ejection Fraction (EF) - the fraction of blood ejected by the ventricle relative to its end-diastolic volume. $EF = (SV / EDV) \times 100$
(Normal values: **50% - 75%**)

Classification of HF based on Ejection fraction:

Heart failure with reduced ejection fraction: Mostly systolic

Heart failure with preserved ejection fraction: mostly diastolic



SYSTOLIC Dysfunction

- Reduced contractility
- Increased afterload
- Reduced stroke volume
- Frank Starling curve shifts down and to the right
- **More common than diastolic failure**

Reduced Ejection Fraction

$$\downarrow EF = (\downarrow SV / \uparrow EDV)$$

DIASTOLIC Dysfunction

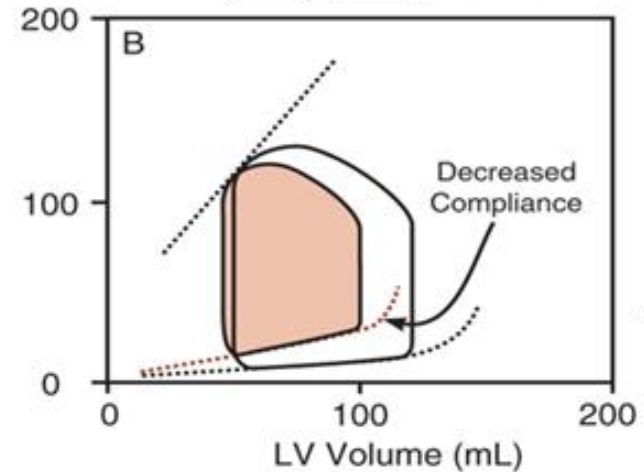
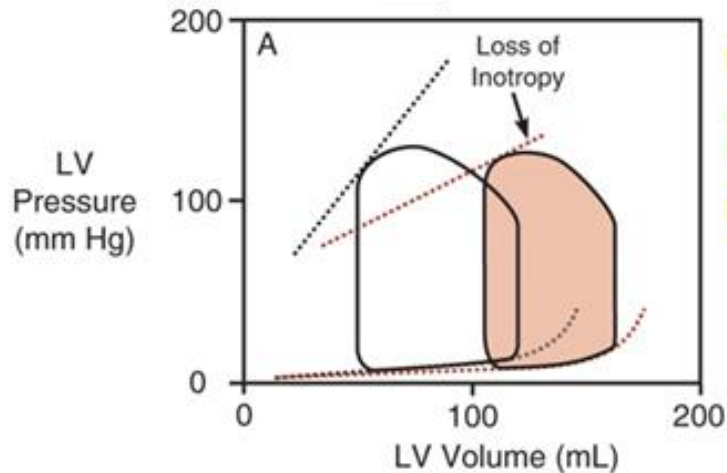
- Failure in filling (mitral stenosis)
- Reduced end diastolic volume
- Reduced stroke volume

Preserved Ejection Fraction

$$\{ \text{Normal range} \} EF = (\downarrow SV / \downarrow EDV)$$

However, there is much overlap between systolic and diastolic failure – many patients demonstrate both types.*

Pressure-Volume Loops in Systolic vs Diastolic Dysfunction



Systolic Dysfunction ($\downarrow$ EF) :

Impaired contractility $\rightarrow$ $\uparrow$ EDV
and passive wall stretch $\downarrow$

Leads to:

- Inc. EDV – increased compliance
- Red. EF
- Inc. blood volume (RAS)
- Inc. wall stress
- Inc. work for heart

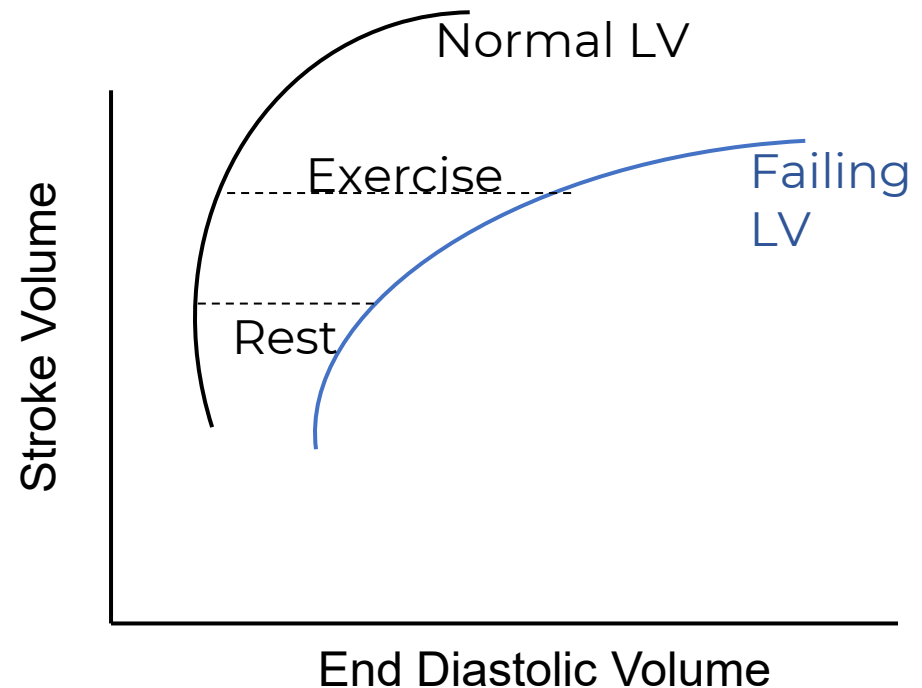
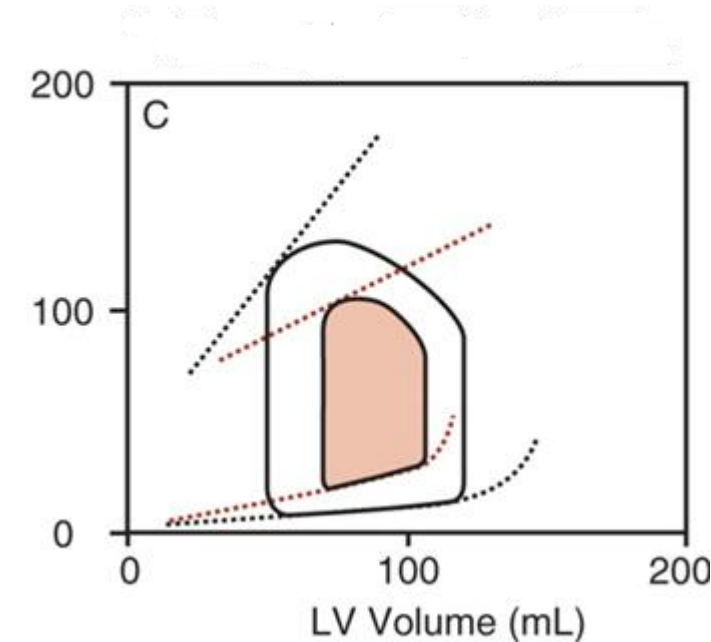
Diastolic Dysfunction

(preserved EF) : Dec.
ventricular compliance with
hypertrophy

Leads to:

- Inc. End Diastolic Pressure,
- Reduced ventricle filling
- Near normal EF
- Reduced SV

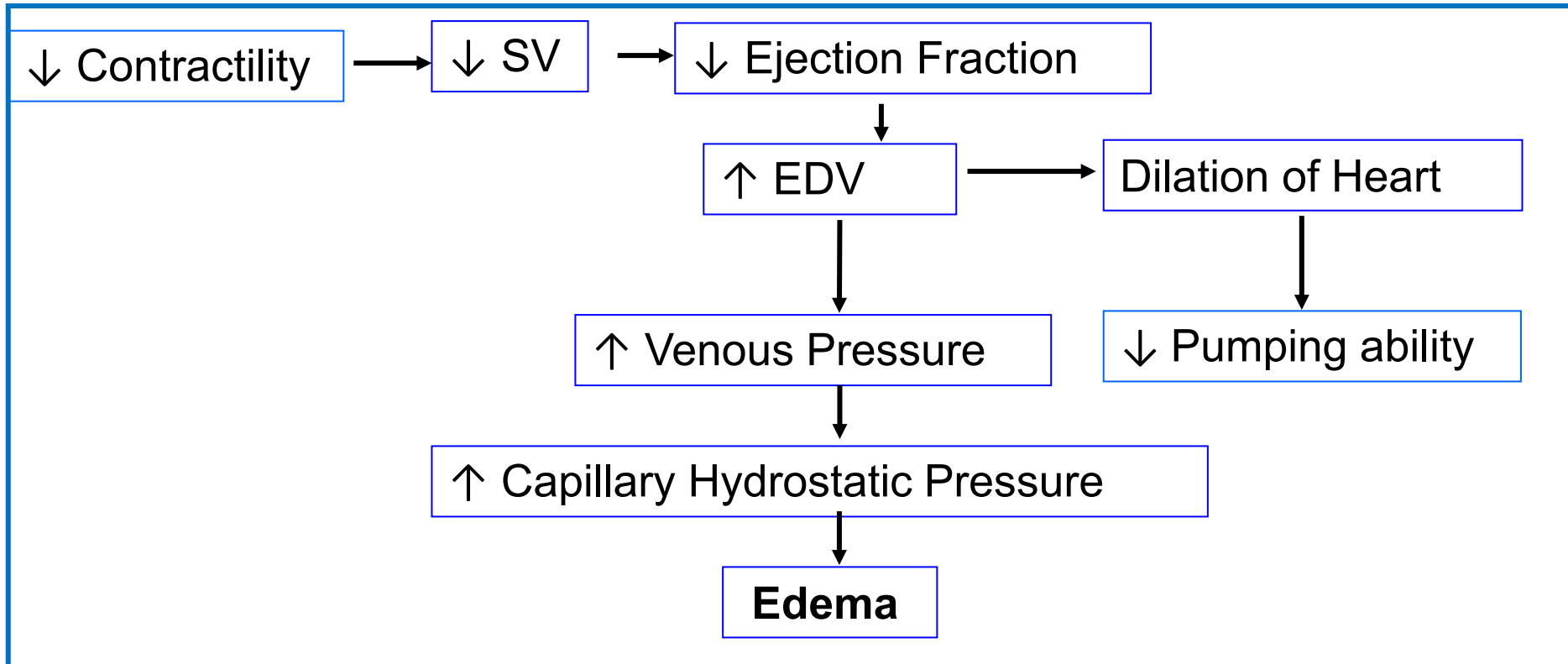
How the Frank-Starling curve (cardiac performance curve) is changed in CHF



Note:

- Compensatory hypertrophy that occurs in heart failure will eventually lead to a reduction in ventricular compliance and reduced filling.
- Manifestations of cardiac failure is more obvious during exercise
- Heart operates on plateau of cardiac performance curve

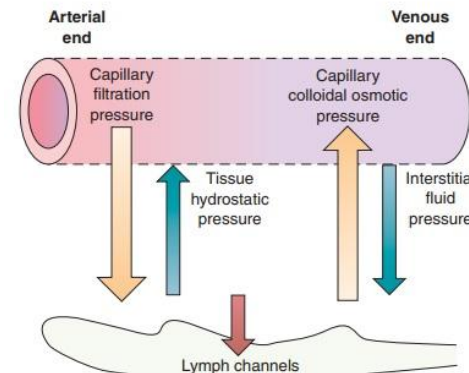
Some Clinical Consequences of ↓ Contractility



• ↑ **Capillary Hydrostatic Pressure** pushes fluid **out** into the interstitium.

• **Plasma Colloid Osmotic Pressure** pulls fluid **back** into the capillary

• *Review microcirculation DLA please 😊



Edema: LV Failure vs RV Failure

LV starts to fail (RV still normal)

↑ LA pressure

↑ Pulmonary venous pressure

Pulmonary edema

Breathlessness
(Dyspnea)

Worse when supine
(Orthopnea)



↑ Pulmonary artery BP (Pulmonary hypertension)

↑ Afterload on RV

RV starts to fail

Peripheral edema

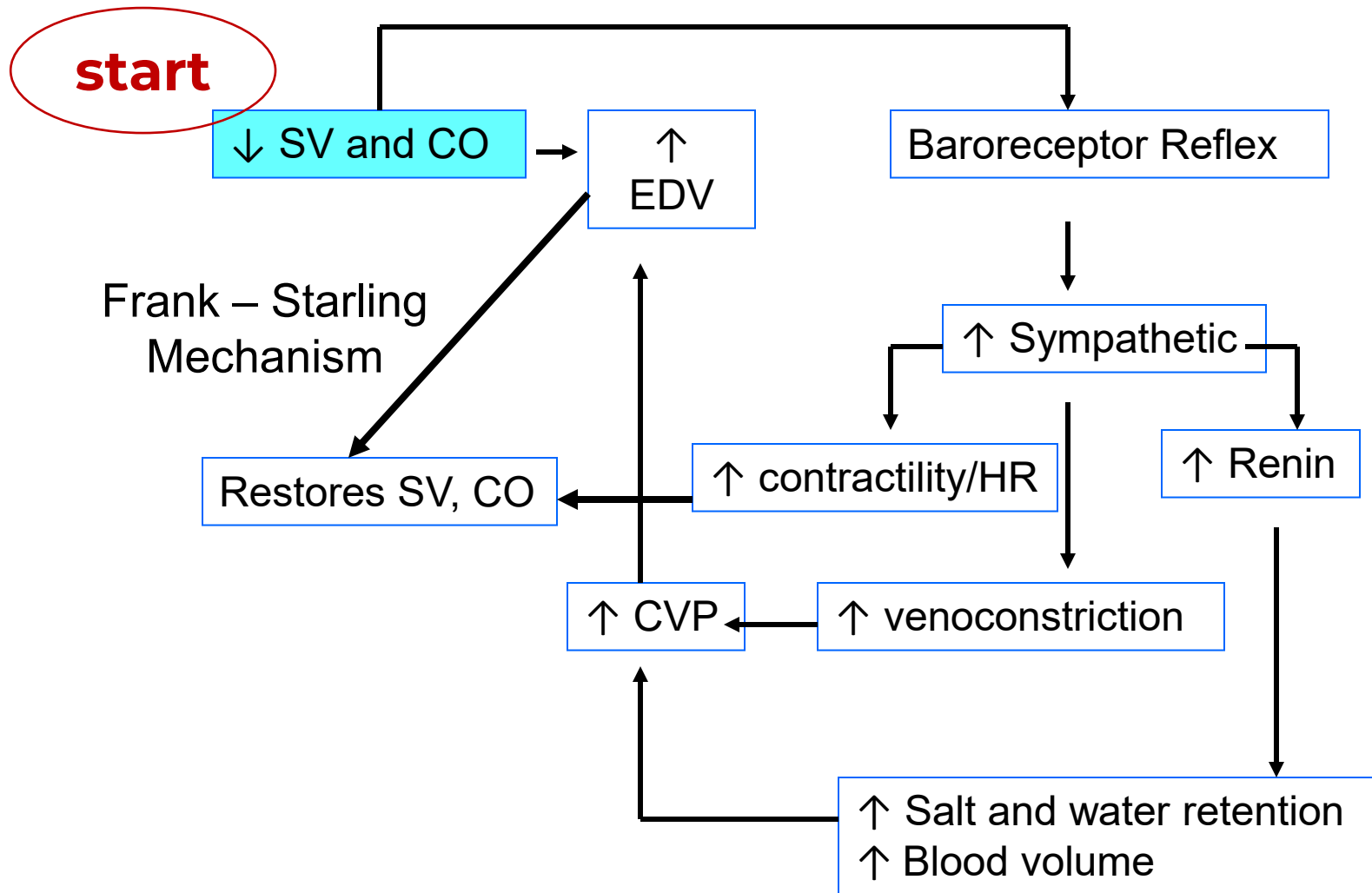
Ankle Swelling

Pitting edema



When one ventricle starts to fail, the other ventricle will eventually fail also.

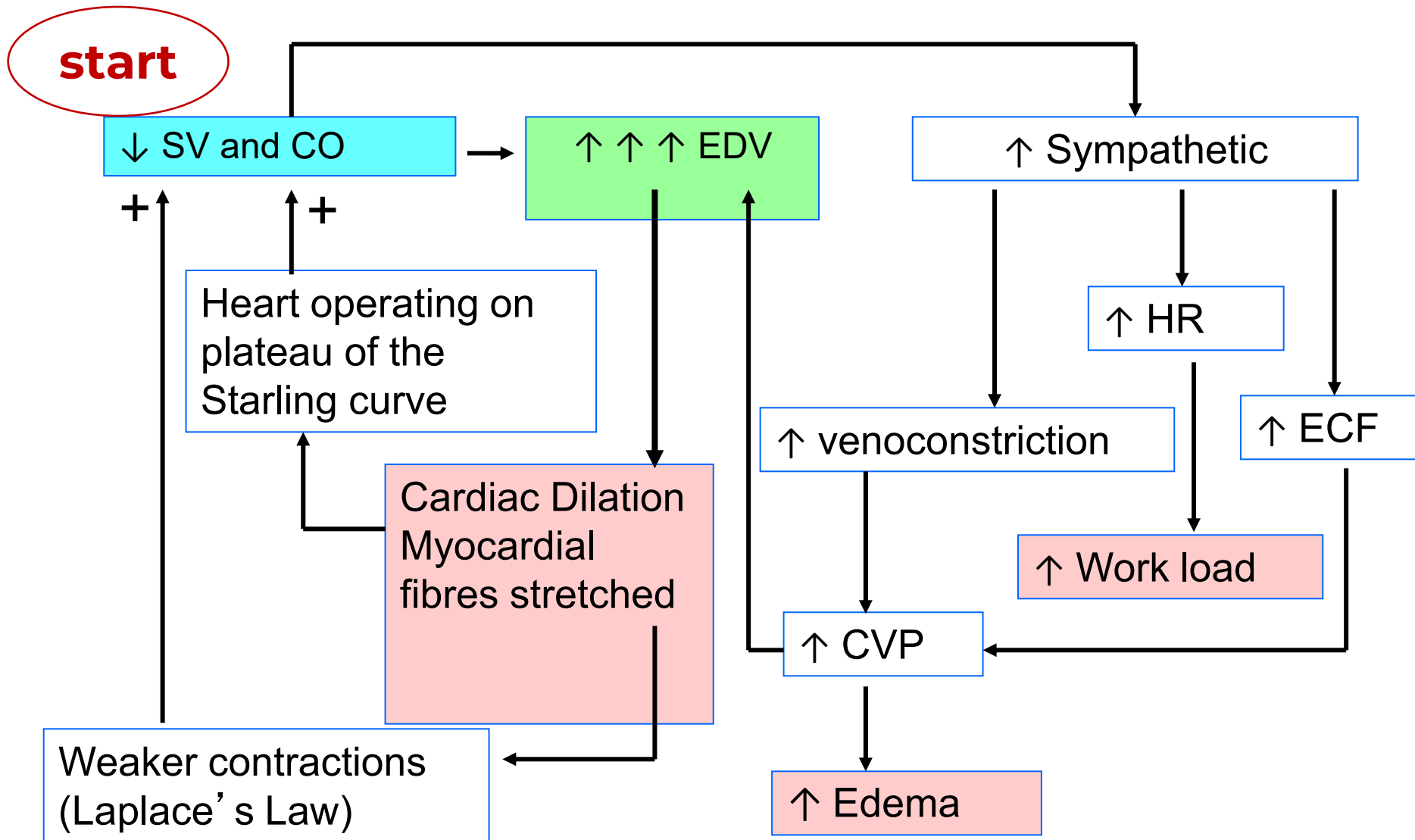
Initial Cardiac Failure: Compensatory Mechanisms



Continued activation of the compensatory mechanisms (initially beneficial) ultimately proves harmful.

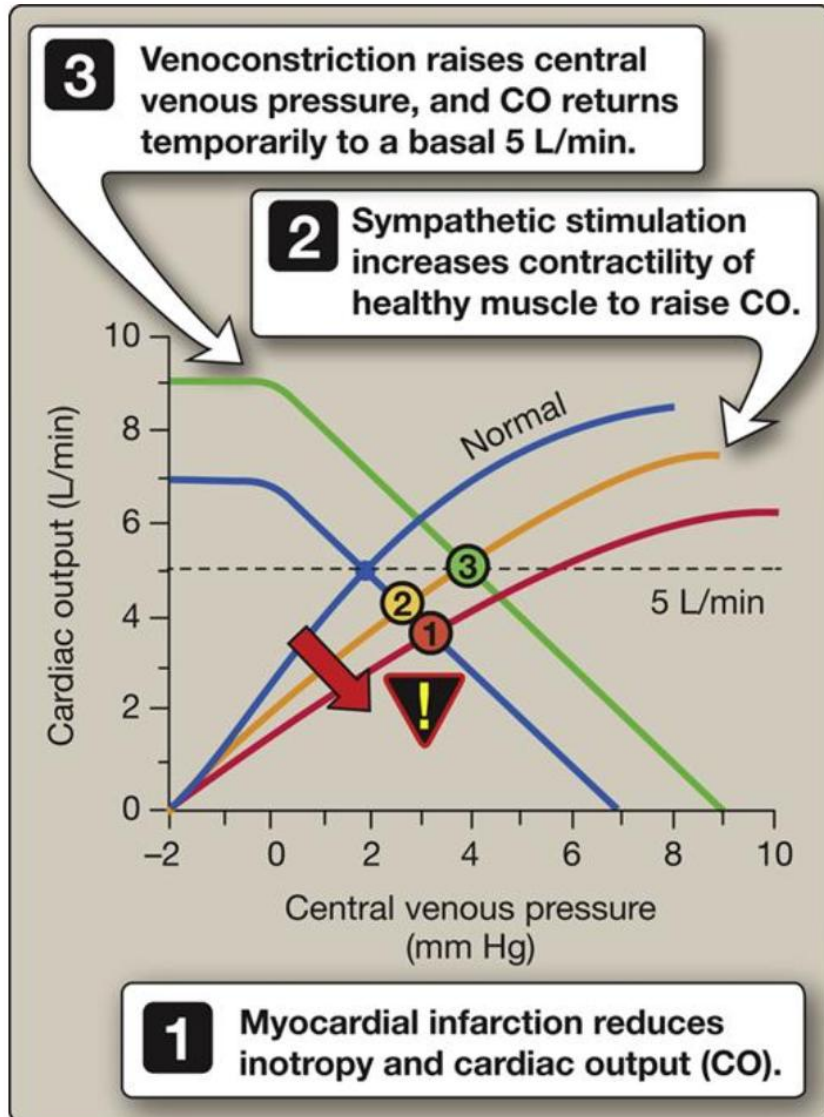
Initial compensatory mechanism	Harmful long-term consequence
↑ Blood volume	Pulmonary edema
↑ TPR	↑ Afterload; ↓ SV and CO
↑ HR	↑ Metabolic demand
Continuous Sympathetic activity	Down regulation of β receptors
↑ Angiotensin II	↑ cytokines and fibroblasts Adverse remodelling of the heart
Heart now goes into decompensated state	

Later Stage of Cardiac Failure: Decompensated Heart

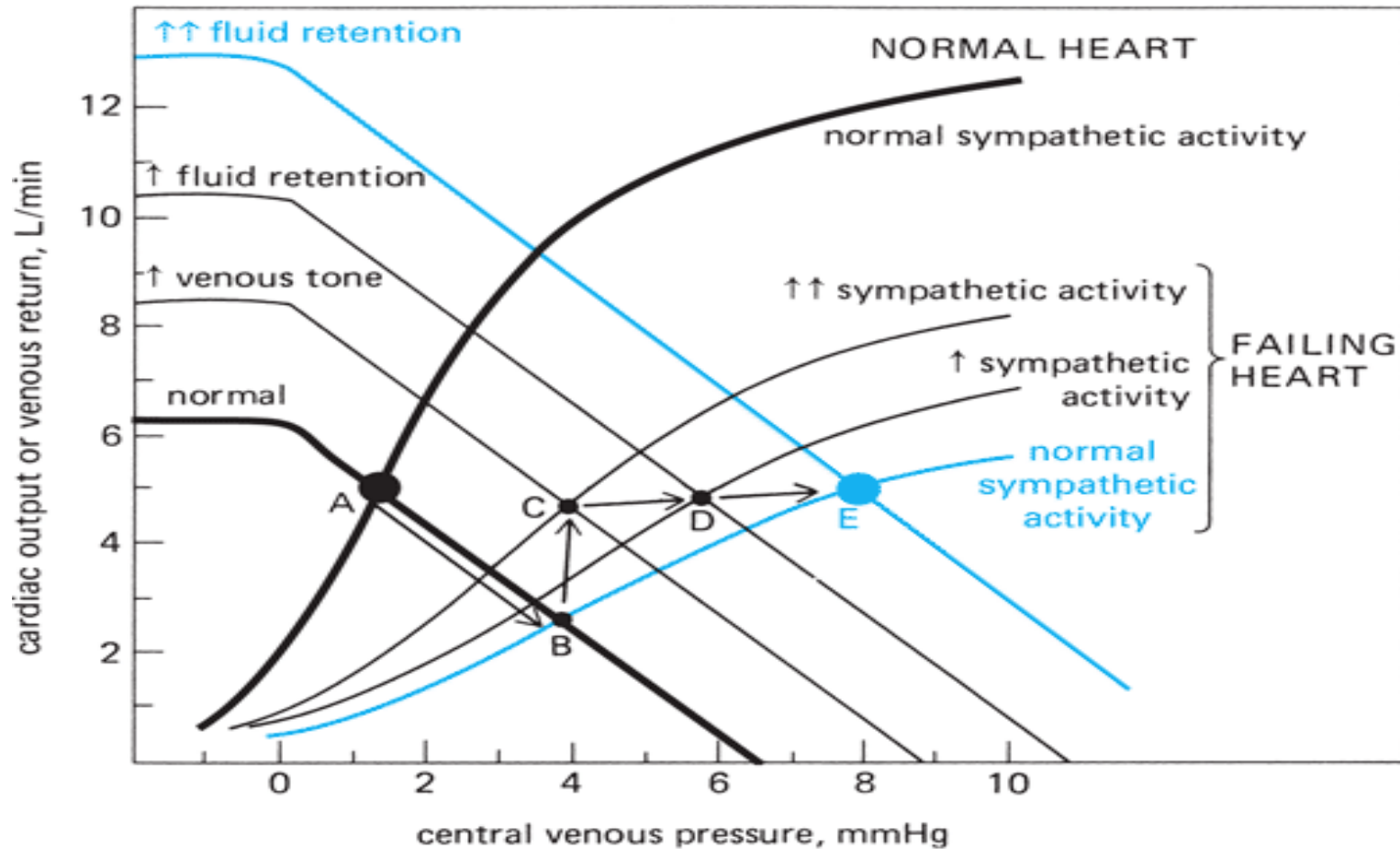


The EDV ↑ is so large – heart now operating on plateau of Frank – Starling curve (compensatory mechanisms become inadequate).

Immediate (Sympathetic) response to reduced cardiac output e.g. caused by MI



Compensated and Decompensated effects of cardiac failure on VR and CO on Guyton Crossplot



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CHF vs Normal Cardiac Response

COMPARISON OF CARDIOVASCULAR FUNCTION IN A NORMAL PERSON AND A PATIENT WITH MODERATE-TO-SEVERE CHF AT REST AND AT MAXIMAL EXERCISE

- Early stages of heart failure - CO and MAP may be normal at rest because of compensatory changes. But the symptoms appear when person performs physical work.
- At rest CHF patient has $\downarrow$ CO caused by $\downarrow$ SV. MAP is slightly $\downarrow$, and resting HR $\uparrow$. Because of lower organ blood flow in CHF, O₂ extraction is $\uparrow$.
- At maximal exercise CHF patient can $\uparrow$ CO by only 50% because of inability of LV to $\uparrow$ SV (plus the lower HR).
- CHF patient has significant $\downarrow$ in MAP because the $\uparrow$ in CO is not sufficient to maintain MAP as the TPR falls during exercise.
- The maximal whole body O₂ consumption is greatly $\downarrow$ in CHF because the reduced perfusion of the active muscles limits O₂ delivery and therefore the O₂ consumption of the muscle.
- CHF patient suffers substantial fatigue and dyspnea during exertion.

Objectives of Treatment of heart failure

- ↓ clinical symptoms of edema and dyspnea
- To improve CV function to enhance organ perfusion and exercise capacity
- ↓ mortality

Achieved by

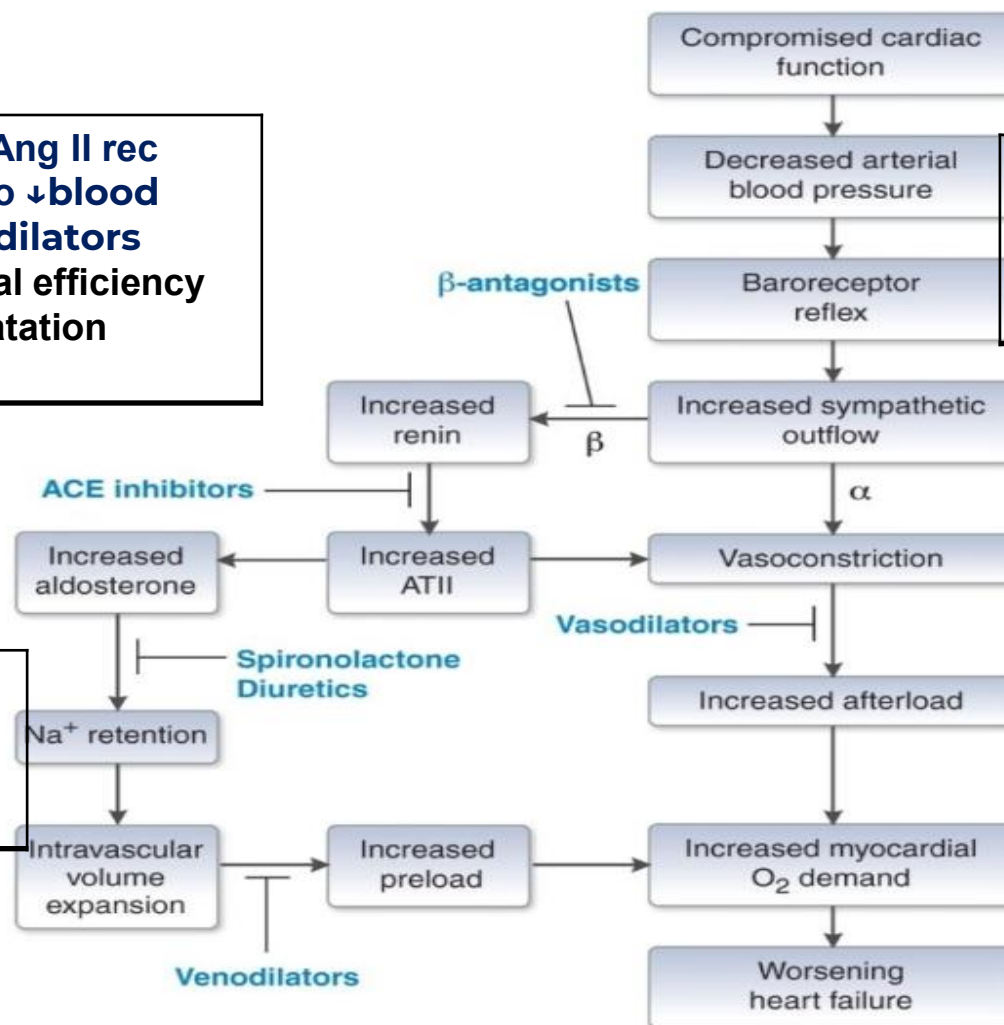
ACE inhibitors, Ang II rec antagonists (also ↓ blood volume), Vasodilators
improve mechanical efficiency
↓ Cardiac dilatation

Beta blockers in small incremental doses

- ↑ SV and CO
- ↓ Afterload

DIURETICS
↓ venous pressure → ↓ edema
↓ Blood volume

DIGOXIN
Control of ventricular rate with a small inotropic effect
↑ Myocardial Contractility



The release of which hormone would be beneficial in heart failure?

Atrial natriuretic peptide

- ↑ Excretion of Na⁺ and water
- Vasodilation
- Inhibit renin secretion
- Antagonize effects of Ang II on aldosterone and ADH

But levels are not usually sufficient to counteract effect of Ang II, aldosterone and ADH

New treatment for heart failure

- Drugs which increase BNP (by blocking its degradation) are now proving useful in management of heart failure.

BNP: B-type (Brain) Natriuretic Peptide

BNP is released from ventricles in response to pressure and volume overload. *Sensitive marker for heart failure.

Clinical Case.

Myocardial Infarction – Left Ventricular Failure

A 52-year-old construction manager is significantly overweight. His rich diet included red meats and high-calorie desserts. He also enjoyed unwinding with a few beers each evening. He joked with the guys, "I guess I'm a heart attack waiting to happen." He had occasional chest pains (angina) that were relieved by nitroglycerin.

The evening of his MI, Marvin went to bed early because he wasn't feeling well. He awakened at 2:00 A.M. with crushing pressure in his chest and pain radiating down his left arm that was not relieved by nitroglycerin. He was nauseated and sweating profusely. He also had difficulty breathing (dyspnea), especially when he was recumbent (orthopnea). His breathing was "noisy." The wife called 911, and paramedics arrived promptly and transported him to the nearest hospital.

Clinical Case: Myocardial Infarction – Left Ventricular Failure

Pulmonary capillary wedge pressure, obtained during cardiac catheterization, was 30 mm Hg (normal, 5 mm Hg). His ejection fraction, measured with two-dimensional echocardiography, was 0.35 (normal, 0.55). (**is this systolic or diastolic dysfunction?*)

He was transferred to the coronary intensive care unit. He was treated with a thrombolytic agent to prevent another myocardial infarction, digitalis (a positive inotropic agent), and furosemide (a loop diuretic). (**why use these drugs?*)

After 7 days in the hospital, he was sent home on a strict, low-fat, low-Na⁺ diet. (**why restrict the diet?*)